

Circuit Elements

Wires, Batteries, Resistors, Capacitors and Inductors

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1 Section 14 — Wires

1.1 Definition

Wires are low-resistance pathways for the flow of electrical currents.

1.2 Additional Information

The diameter of a wire is expressed by a quantity called **gauge number**.

Larger wire diameter → Smaller gauge number → Smaller resistance

1.3 Types of Wires

Based on Forms

- **Solid Core** — a single blob of conductor
- **Stranded Wire** — multiple strands of conductor (for instance Cu)
- **Braided Wire** — multiple strands of conductors are braided together

Based on Applications

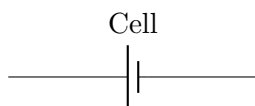
- **Pre-tinned solid bus wire** — for hobbyists and Printed Circuit Board (PCB) applications
- **Speaker wire** — stranded together for higher current flow
- **Magnet wire** — for building magnetic cores

2 Section 15 — Batteries

2.1 Definition

A battery is a source of energy that converts chemical energy into electrical energy and induces current flow in circuits.

Cell: A fundamental unit of energy storage in a battery that converts chemical energy to electrical energy through oxidation–reduction reactions involving exchange of electrons. A full battery can be just one cell or many cells linked together.



Components of a Cell

- **Negative Terminal / Electrode** — Anode — a source for electrons
- **Positive Terminal / Electrode** — Cathode — a sink for electrons
- **Electrolyte** — liquid or paste-like — a transporter of charge carrying particles (electrons or ions)

Example of a Cell — The Lead-Acid Cell

- **Anode:** Lead, $\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$
- **Electrolyte:** Sulfuric acid solution $(\text{H}_2\text{O} + \text{H}_2\text{SO}_4 \rightarrow \underbrace{3\text{H}^+ + \text{SO}_4^{2-} + \text{OH}^-}_{\text{Charge Carriers}})$
- **Cathode:** Lead oxide, $\text{PbO}_2 \rightarrow \text{Pb}^{2+} + \text{O}_2^{2-}$
- **Overall Reactions:**
 - Anode — $\text{Pb} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4 + 2\text{e}^-$
 - Cathode — $\text{PbO}_2 + \text{SO}_4^{2-} + 4\text{H}^+ \rightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$

Typical Voltage across a Cell: 1.5 V

2.2 Types of Batteries

Based on Longevity of “Cells”

1. **Primary Batteries** — designed for one-time use
 - Carbon-zinc batteries
 - Alkaline batteries
2. **Secondary Batteries** — rechargeable batteries
 - Nickel metal-hydride batteries
 - Lead-acid batteries

2.3 Battery Capacity

A quantity that determines how much electrical energy a battery can provide before being fully drained. It is measured in ampere-hours (Ah).

Higher Capacity $\rightarrow$ Longer lifetime

Time of operation before being fully drained:

$$t = \frac{\text{Battery Capacity}}{\text{Current}}$$

Typical battery capacities:

AAA $\rightarrow$ 1000 mAh

AA $\rightarrow$ 2300 mAh

C $\rightarrow$ 5000 mAh

D $\rightarrow$ 8500 mAh

9-V NiMH $\rightarrow$ 250 mAh

2.4 C Rating

A quantity that determines charge and discharge current and time.

Higher C Rating $\rightarrow$ Higher Current & Lower discharge time

C Rating is determined from the actual time to discharge completely or the current flow:

$$\text{C Rating} = \frac{\text{Rated Capacity}}{\text{Time to discharge} \times \text{Rated Current}} = \frac{\text{Current} \times 1 \text{ hr}}{\text{Rated Capacity}}$$

C Rating depends primarily on **internal resistance**.

3 Section 16 — Resistors

3.1 Definition

Resistors are circuit elements that **control the current flow through them and thereby the current flow in the circuit**.

3.2 Additional Information

Resistors have primarily two major functions:

1. Limiting the current flow.
2. Maintaining the voltage level across terminals in circuits.

Controlling the current flow and setting the voltage levels is useful for different kinds of electrical and electronic circuit components. For example:

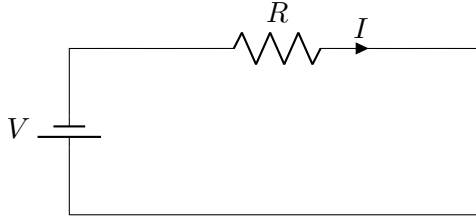
- Controlling the **brightness of an LED circuit**.
- Controlling the **voltage level across a circuit element**.
- The **voltage division microcontroller input circuit**.
- Setting operating current and signal levels in circuits.
- Behaving like damping agents in oscillator circuits.

3.3 Resistance and Ohm's Law

Ohm's Law states that the current flow across a resistor is linearly proportional to the voltage applied across its terminals.

The constant of proportionality that relates current flow to voltage is defined as the inverse of the resistance of the resistor.

Higher Resistance $\rightarrow$ Lower Current Flow (for the same voltage)



Ohm's Law:

$$I = \frac{1}{R} \cdot V$$

where I is the current flow in Amperes (A), V is the voltage in Volts (V) and R is the resistance in ohms (Ω).

3.3.1 Power Loss in Resistors

The simplest expression of power consumed by a resistor occurs for the DC voltage case, where it is given by

$$P = IV = \frac{V^2}{R} = I^2 R$$

where P is the power consumed in Watts (W), I is the current flow in Amperes (A), R is the resistance in Ohms (Ω) and V is the voltage across the resistor (V).

Power Rating: The power rating of a resistor is defined as the maximum amount of power that a resistor can consume without being damaged. It essentially acts as a threshold on the total power that can be dissipated across the resistor.

A resistor should have a power rating of at least twice the maximum power loss anticipated across it.

3.4 Resistor Combinations

Resistors can be combined in circuits to create a complex network of resistors depending upon the application. On a more fundamental level however, combination of resistors can be done in one of two ways:

1. Parallel Combination
2. Series Combination

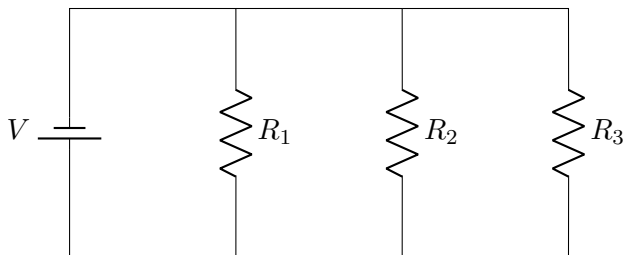
If we swap out an arbitrary combination of resistors with just a single resistor such that the current through it and the voltage across it remain the same as in the arbitrary network of resistors, then the resistance of that single resistor is called the **equivalent resistance** for that resistor combination. We can think of the equivalent resistance as a way of simplifying a rather complex combination of resistors into just a single resistor.

3.4.1 Parallel Combination of Resistors

When multiple resistors are connected between two common nodes they are connected in a parallel configuration.

For a parallel configuration of resistors, the current has multiple parallel paths to flow through. As a result, the equivalent resistance of a parallel configuration of resistors is lower than (even the smallest of) the individual resistors' resistance.

- **Voltage across** all parallel resistors is the same.
- **Total current flow** through a set of parallel resistors is divided among the parallel resistors and is inversely proportional to the resistance value ($I \propto \frac{1}{R}$).



Thus, for a parallel configuration of resistors, the reciprocal of the equivalent resistance is equal to the sum of the reciprocals of all the constituent resistances.

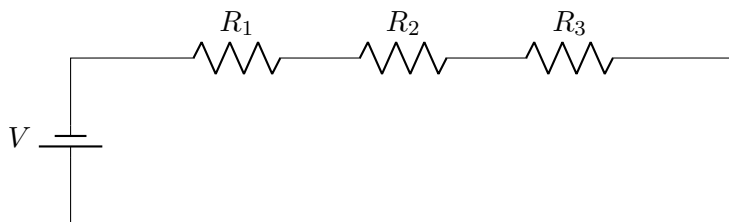
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

where R_{eq} is the equivalent resistance of the parallel combination and $R_1, R_2, \dots, R_n$ are the individual resistances.

3.4.2 Series Combination of Resistors

Resistors are said to be connected in series if the ending terminal of one resistor is connected to the starting terminal of the next resistor, and so on.

- **Total voltage across** multiple resistors in series is divided across each resistor in proportion to their resistance values.
- **Current flow** is equal across all resistors.



The total equivalent resistance of multiple resistors in series is given by the following formula.

$$R_{eq} = R_1 + R_2 + \cdots + R_n$$

where R_{eq} is the equivalent resistance of the series combination and $R_1, R_2, \dots, R_n$ are the individual resistances.

3.5 Resistor Types

3.5.1 Based on How Resistances Are Presented

Physically, resistors can be of two kinds based on how their resistances are branded on them:

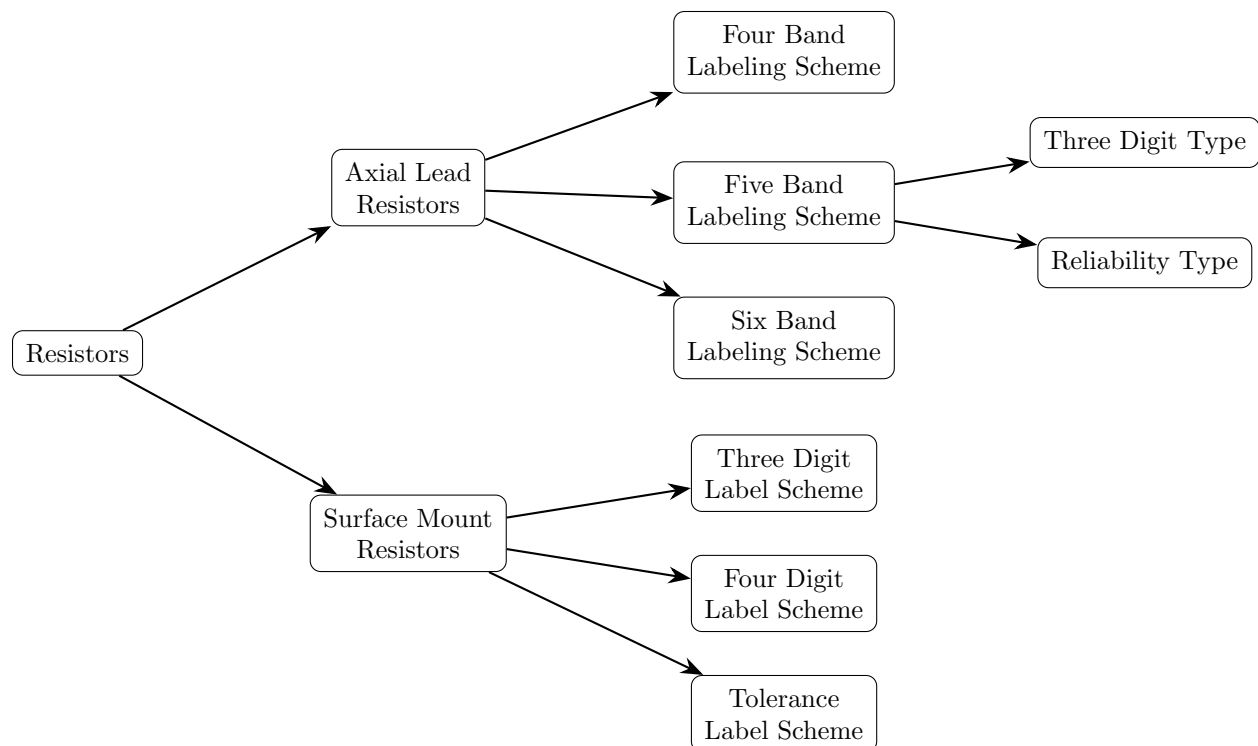
1. Axial lead resistors
2. Surface mount resistors

Axial lead resistors can have the resistance value labeled on them using at least three different schemes:

1. Four band labeling scheme
2. Five band labeling scheme
3. Six band labeling scheme

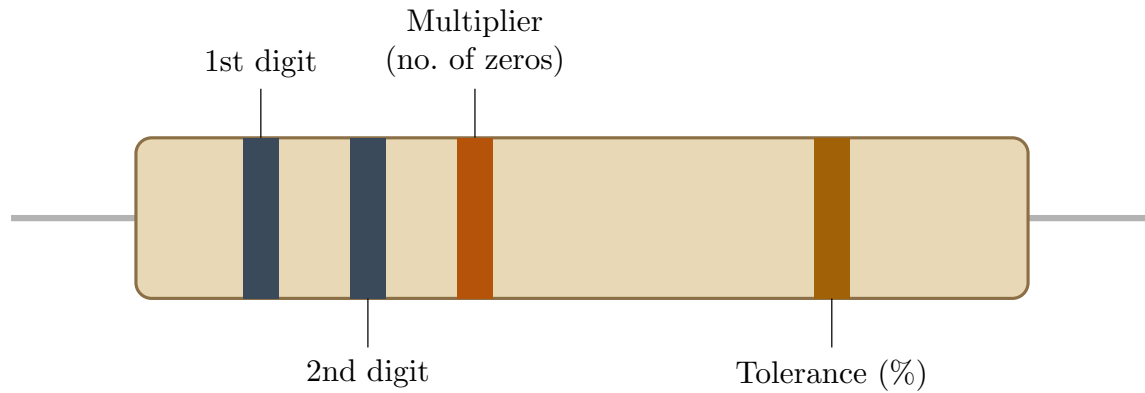
Surface mount resistors can be labeled using three different schemes:

1. Three digit label scheme
2. Four digit label scheme
3. Tolerance label scheme

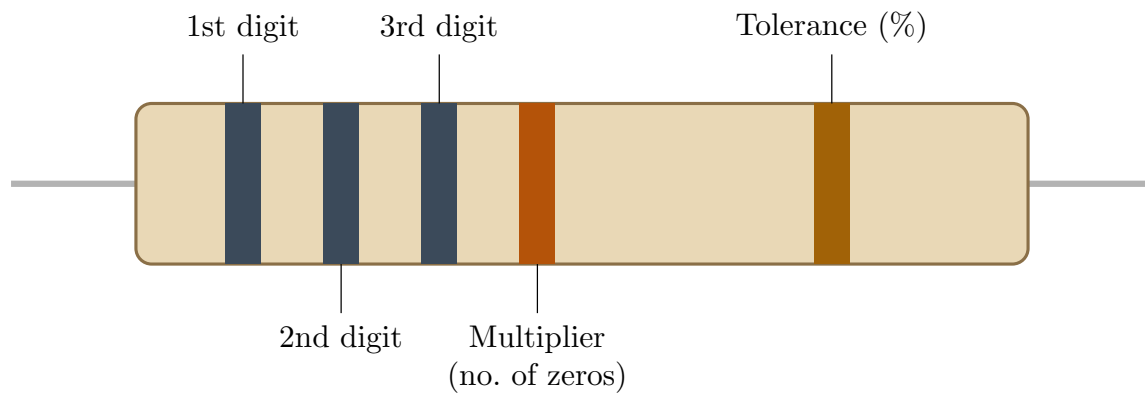


3.5.2 Interpreting the Axial Lead Resistor Labels

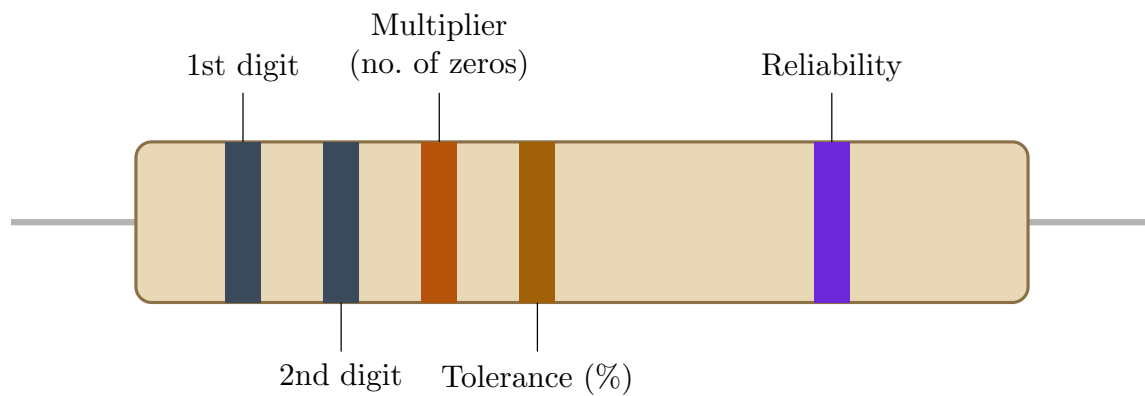
Four Band Labeling Scheme



Five Band Labeling Scheme — Three Digit Type

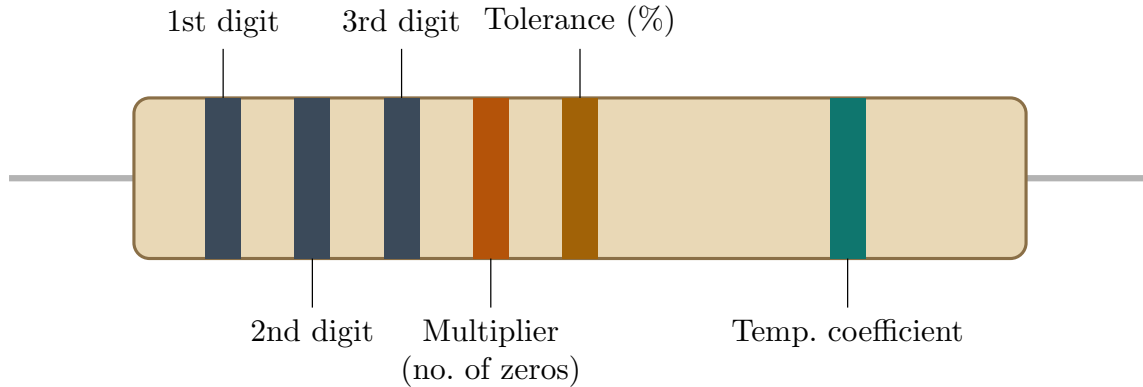


Five Band Labeling Scheme — Reliability Type



Reliability denotes the percentage change in resistance over a fixed time interval, usually taken as 1000 hours.

Six Band Labeling Scheme

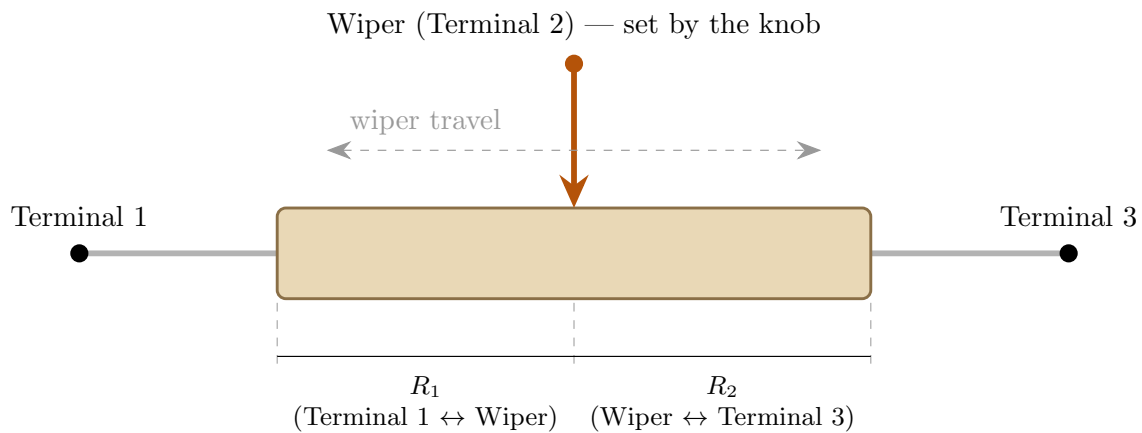


3.5.3 Based on the Changeability of the Resistance Value

Resistors can be of two types based on whether the resistance values can be changed or not.

1. Static / Fixed Resistors
2. Variable Resistors (Rheostat / Potentiometer / Trimmers)

Potentiometers, also called *pots* for short, are a kind of variable resistor with three terminals which usually has a rotation knob to control the resistance value in between the peripheral (end) terminals and the center one.



Rotating the knob moves the wiper along the resistive track, which splits the fixed end-to-end resistance into two parts. The total resistance between the two end terminals stays constant:

$$R_1 + R_2 = R_{13} = \text{constant}$$

Potentiometers are further classified based on how frequently one needs to adjust the voltage in a pot.

- **Preset pot / Trimpot** — resistance values are only adjusted occasionally.
- Pots that handle frequent resistance adjustments.

The control knob of a potentiometer can map to the resistance across its two terminals in different ways. The change of total resistance versus the effective rotation can be of the following kinds:

1. Linear taper
2. Log taper
3. True logarithmic taper

3.6 Adjusting Current and Voltage with Variable Resistors

3.6.1 Adjusting Current

$$I = \frac{V_{\text{IN}}}{R_{\text{pot}} + R_{\text{load}}}$$

3.6.2 Adjusting Load Voltage

$$V_{\text{L}} = \frac{R_{\text{L}}}{R_{\text{L}} + R_{\text{pot}}} \cdot V_{\text{IN}}$$

3.7 (Bonus) The Y- Δ Transformation

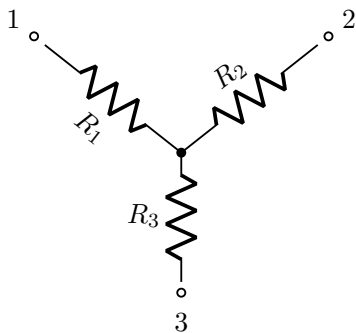
Not every resistor network can be reduced using only series and parallel combinations. Some networks, such as a bridge network, contain resistors arranged in a **Y** (wye, also called a star or T) configuration or a Δ (delta, also called a pi) configuration, which cannot be simplified any further by the series-parallel rules alone.

The **Y- Δ transformation** lets us convert between these two configurations. Once converted, the rest of the network can usually be reduced by the ordinary series and parallel rules.

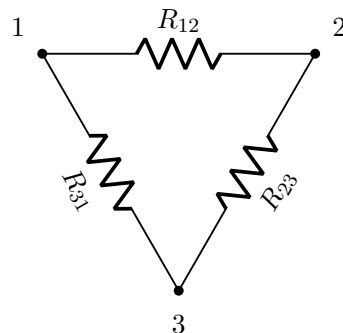
Both configurations connect the same three terminals, which we label 1, 2 and 3.

- In the **Y** configuration, the three resistors R_1 , R_2 and R_3 each connect one terminal to a common central node.
- In the Δ configuration, the three resistors R_{12} , R_{23} and R_{31} connect the terminals directly to each other in a loop.

Two configurations are equivalent if the resistance measured between any pair of terminals is the same in both.



Y (wye) configuration



Δ (delta) configuration

3.7.1 $\Delta \rightarrow Y$ Conversion

Each Y resistor is the product of the two adjacent Δ resistors divided by the sum of all three Δ resistors.

$$R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

$$R_2 = \frac{R_{12} \cdot R_{23}}{R_{12} + R_{23} + R_{31}}$$

$$R_3 = \frac{R_{23} \cdot R_{31}}{R_{12} + R_{23} + R_{31}}$$

3.7.2 $Y \rightarrow \Delta$ Conversion

Each Δ resistor is the sum of the pairwise products of the Y resistors, divided by the Y resistor opposite to it.

$$R_{12} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$$R_{23} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_{31} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

3.7.3 Special Case: All Resistors Equal (Balanced Network)

If all three resistors in a configuration are equal, the formulas simplify considerably.

If all the Δ resistors are equal to R_Δ , then each Y resistor becomes

$$R_Y = \frac{R_\Delta \cdot R_\Delta}{3R_\Delta} = \frac{R_\Delta}{3}$$

Conversely, if all the Y resistors are equal to R_Y , then each Δ resistor becomes

$$R_\Delta = \frac{3R_Y^2}{R_Y} = 3R_Y$$

So for a balanced network the two are related by the simple factor of three:

$$R_\Delta = 3R_Y \quad \Longleftrightarrow \quad R_Y = \frac{R_\Delta}{3}$$

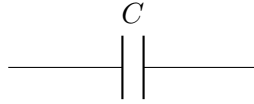
Balanced $\Delta \rightarrow$ divide by 3 $\rightarrow$ Balanced Y
--

Balanced Y $\rightarrow$ multiply by 3 $\rightarrow$ Balanced Δ
--

4 Section 17 — Capacitors

4.1 Definition

Capacitors are energy storage elements that store energy within the electrostatic field inside them.



4.2 Additional Information

Capacitors perform two major functions:

1. Store energy for a short amount of time.
2. Isolate AC quantities from DC quantities — a capacitor **blocks DC** and **allows AC** to pass through.

By default, a capacitor prevents sharp changes in the voltage across its terminals. As a result, it can smooth out ripples in voltages and also cause a slow change in voltage.

4.3 Capacitance

Capacitance is defined as the amount of charge stored in a capacitor for a unit change of voltage across its terminals. It is given by the equation shown below.

$$C = \frac{Q}{V}$$

where C is the capacitance in Farads (F), Q is the charge stored within the capacitor in Coulombs (C) and V is the voltage across the capacitor terminals in Volts (V).

For instance, if a capacitor has a capacitance of 1 pF, then it will store 1 pC (10^{-12} C) of charge for an applied voltage of 1 V across its terminals.

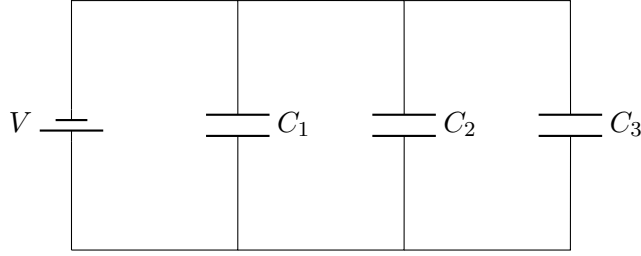
4.4 Combination of Capacitors

Similar to resistors, capacitors can be combined in circuits in any arbitrary combination. If we swap out an arbitrary combination of capacitors with just a single capacitor such that the charge stored by it and the voltage across it remain the same as in the arbitrary network of capacitors, then the capacitance of that single capacitor is called the **equivalent capacitance** for that capacitor combination. We can think of the equivalent capacitance as a way of simplifying a rather complex combination of capacitors into just a single capacitor. However, on a more fundamental level, capacitors can have either a series combination or a parallel combination.

1. Parallel Combination
2. Series Combination

4.4.1 Capacitors in Parallel Combination

When multiple capacitors are connected between the same two terminals, they are said to be in a parallel combination.



The total capacitance of multiple capacitors in parallel is just the sum of the individual capacitances.

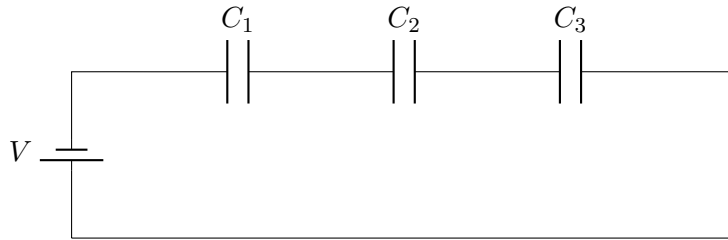
$$C_{eq} = C_1 + C_2 + \cdots + C_n$$

where C_{eq} is the equivalent capacitance of the parallel combination and $C_1, C_2, \dots, C_n$ are the individual capacitances.

When multiple capacitors are connected in parallel, the voltage across them remains constant, but the total charge stored is divided among the different capacitors in proportion to their capacitance. Thus, the total (equivalent) capacitance is the sum of the individual capacitances.

4.4.2 Capacitors in Series Combination

When multiple capacitors are connected in such a way that the end terminal of one capacitor is connected to the starting terminal of the next capacitor, and so on, then the capacitors are said to be in a series combination.



The reciprocal of the total capacitance of multiple capacitors in series is just the sum of the reciprocals of the individual capacitances.

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

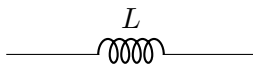
where C_{eq} is the equivalent capacitance of the series combination and $C_1, C_2, \dots, C_n$ are the individual capacitances.

When multiple capacitors are connected in series, the charge stored across all of them remains constant, whereas the voltage is divided among the different capacitors in inverse proportion to their capacitances. This is why the equivalent capacitance takes the form given above.

5 Section 18 — Inductors

5.1 Definition

Inductors are another kind of energy storage element that stores energy within the **magnetic field** inside it, unlike the capacitor which stores energy in the electric field inside it.



5.2 Additional Information

Just as a capacitor prevents abrupt changes in voltage across its terminals, an inductor prevents abrupt changes in current across its terminals.

Just as a capacitor blocks DC voltage but allows AC voltage, an inductor **blocks DC current** but **allows AC current** to flow.

An inductor blocks high frequency signals but allows low frequency signals to pass — the opposite of how a capacitor works.

So when used in tandem with a capacitor, an inductor and a capacitor together can create **resonance circuits** — circuits that generate a sinusoidally varying voltage. Additionally, a combination of inductors with resistors can create either a **low pass filter** or a **high pass filter**.

5.3 Inductance

Inductance is defined from the property of an inductor to induce a voltage depending on the changing current through the coil. It is defined as the amount of induced voltage that opposes the change of current, for a unit change of current over time. It is given by the equation shown below.

$$V = -L \frac{dI_L}{dt}$$

where V is the induced voltage across the inductor acting in opposition to the direction of current flow defined by I_L , L is the inductance in Henries (H), and $\frac{dI_L}{dt}$ denotes the rate of change of current over time in Amperes per second (A/s).

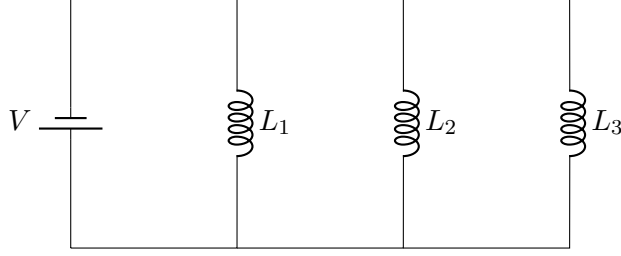
5.4 Combination of Inductors

Similar to resistors and capacitors, inductors can be combined in circuits in any arbitrary combination. If we swap out an arbitrary combination of inductors with just a single inductor such that the induced voltage across it and the current through it remain the same as in the arbitrary network of inductors, then the inductance of that single inductor is called the **equivalent inductance** for that inductor combination. We can think of the equivalent inductance as a way of simplifying a rather complex combination of inductors into just a single inductor. On a more fundamental level, inductors can have either a series combination or a parallel combination.

1. Parallel Combination
2. Series Combination

5.4.1 Inductors in Parallel Combination

When multiple inductors are connected between the same two terminals, they are said to be in a parallel combination.



The reciprocal of the total inductance of multiple inductors in parallel is just the sum of the reciprocals of the individual inductances.

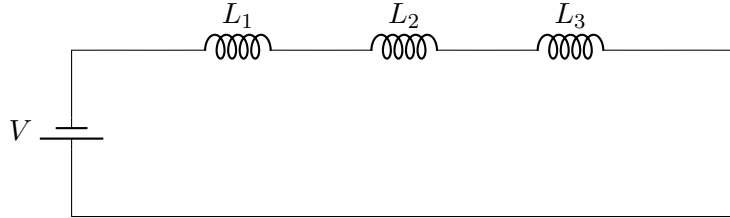
$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_n}$$

where L_{eq} is the equivalent inductance of the parallel combination and $L_1, L_2, \dots, L_n$ are the individual inductances.

When multiple inductors are connected in parallel, the voltage across them remains constant, but the total current is divided among the different inductors in inverse proportion to their inductance. This is why the equivalent inductance takes the form given above.

5.4.2 Inductors in Series Combination

When multiple inductors are connected in such a way that the end terminal of one inductor is connected to the starting terminal of the next inductor, and so on, then the inductors are said to be in a series combination.



The total inductance of multiple inductors in series is just the sum of the individual inductances.

$$L_{eq} = L_1 + L_2 + \cdots + L_n$$

where L_{eq} is the equivalent inductance of the series combination and $L_1, L_2, \dots, L_n$ are the individual inductances.

When multiple inductors are connected in series, the current through all of them remains constant, whereas the induced voltage is divided among the different inductors in proportion to their inductance. Thus, the total (equivalent) inductance is the sum of the individual inductances.